

MUKUND KOMATI

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EDUCATION

INDIANA UNIVERSITY

M.S. in Data Science, Big Data Specialization

Bloomington, IN, USA

AUGUST 2024-MAY 2026

EXPERIENCE

FHL BANK INDIANAPOLIS

Data Platform Engineering Intern

Indianapolis, IN, USA

JUNE 2025-PRESENT

Modernize Bank's Data Infra by migrating on-prem systems to Azure (Azure | Fabric | PySpark | Terraform):

- Built a metadata-driven Fabric ingestion framework to pull on-prem SQL tables into Lakehouse with dynamic table discovery, schema typing, and row-count validation into Delta tables.
- Developed reusable Terraform modules for Fabric workspaces and core data services, standardizing analytics environments across dev, test, and prod.
- Owned database deployments across on-prem SQL and Fabric, coordinating with analytics and application developers to deliver consistent, auditable releases.

DEPT. OF STATISTICS, INDIANA UNIVERSITY

Rust Developer (Machine Learning)

Bloomington, IN, USA

AUGUST 2024-PRESENT

Bayesian Modeling Framework Migration & Optimization (C++|Rust):

- Migrated the Bayesian tree modeling framework (OpenBT) from C++ to Rust to enhance memory safety and performance.
- Reduced memory-related bugs by 70% and improved computational speed by 30%, enabling faster and more reliable Bayesian inference on large datasets.
- Currently testing Rust-based Bayesian tree models using Zero-Knowledge Proofs (ZKPs) to ensure the training process is secure, trustworthy, and repeatable.

PORTER

Machine Learning Engineer

Bengaluru, India

JULY 2023-JULY 2024

Improve driver-customer matchmaking efficiency using a ranking model (Snowpark|Kafka|Flink|Streamlit):

- Architected a centralized Feature Store (60+ engineered features) supporting multiple ML models and automated retraining workflows through Airflow and Snowpark.
- Developed and tuned ranking and classification models using scikit-learn, XGBoost, and PyTorch, optimizing match scores for driver-order pairing and improving the order-fulfillment rate by 4%.
- Conducted A/B tests on over 400k daily orders to compare model versions, validating key order matchmaking KPIs and supporting a safe and effective product rollout.

PORTER

Data Engineer

Bengaluru, India

APRIL 2022-JUNE 2023

Flag fraudulent practices among drivers through the analysis of driver location trajectories (PySpark|Redis|Airflow|Power BI):

- Built and deployed a heuristic model for fraud detection with a precision of 87%.
- Orchestrated a comprehensive ETL project that processed large-scale location data from Redis using PySpark & Airflow.
- Delivered organization-wide dashboards using Power BI and Streamlit to enhance fraud monitoring across millions of daily orders, enabling 3x faster decision-making across departments.
- Acquired expertise in analyzing Geospatial Datasets using PostgreSQL extensions like PostGIS.

BRIDGEI2I (ACCENTURE AI)

Analytics Consultant

Bengaluru, India

JULY 2020-MARCH 2022

Evaluate effectiveness and performance of different marketing activities (Ads & promotions) for a product (R|Python):

- Designed linear and hierarchical Market Mix Models (glm, lmer) in R to quantify the impact of media spend on overall sales, through Ad-ROI and campaign effectiveness.
- Implemented Bayesian models (rstanarm, brms) with domain-informed priors to stabilize parameter estimates 30% faster.
- Derived saturation curves from Marketing Mix Models using Python (Pandas, Matplotlib), identifying optimal Ad spend thresholds and increasing marketing ROI by 15-20% through data-driven budget optimization.

TECHNICAL SKILLS

Languages: Python, C++, Rust, R

ML & Stats: scikit-learn, XGBoost, TensorFlow/Keras, PyMC/Stan, Bayesian Modeling, Causal Inference, Experimentation

Databases: Oracle, MS SQL Server, PostgreSQL, Snowflake SQL, Redis

Big Data: Hadoop, Spark, Kafka, Flink

Data Visualization: Streamlit, Metabase, RShiny, Power BI, Matplotlib, Seaborn, Plotly

Cloud & Deployment: Azure (Data Factory, Fabric, DevOps, ADLS, Monitor, Key Vault), AWS (EMR, EC2, Sagemaker, S3, Glue, Lambda, Athena, Redshift), Airflow, Docker, Jenkins, Terraform, MLflow, Kubeflow

PROJECTS

Semantic Book Recommendation System (Python, LLM Embeddings, ChromaDB, HuggingFace, Gradio):

- Built a semantic recommendation engine using LLM embeddings, improving retrieval relevance 3.2x over TF-IDF on a 100K-book dataset.
- Enhanced category prediction accuracy from 72% (baseline logistic regression) → 86% using zero-shot classification (BART-MNLI).
- Added DistilRoBERTa-based emotion scoring to enable tone-aware recommendations, boosting pilot CTR by 22%.
- Developed a full pipeline for data cleaning, embeddings, vector indexing, and similarity search with ChromaDB.
- Deployed an interactive Gradio UI supporting natural-language queries for real-time semantic recommendations.

Real-Time Stock Market Data Analysis (Kafka|AWS Glue|AWS Athena):

- Engineered a real-time stock analysis system using Apache Kafka for streaming live market data.
- Leveraged AWS S3, Glue, and Athena to enable scalable data storage, cataloging, and efficient querying of large historical and real-time datasets for analysis and reporting.